

```

Fedora release 7 (Moonshine)
Kernel 2.6.35-rc3-next-20100621-dirty on an x86_64

localhost login: root
Last login: Thu Jul 8 03:50:35 on tty0
[root@localhost ~]# lsmod
Module                  Size      Used by
binfmt_misc             6591      1
[root@localhost ~]# modprobe loop
[root@localhost ~]# lsmod
Module                  Size      Used by
loop                    11975      0
binfmt_misc             6591      1
[root@localhost ~]# cat /sys/module/loop/sections/.text
0x7187c000
[root@localhost ~]# dd if=/dev/zero of=fs.img bs=1024K count=2
2+0 records in
2+0 records out
2097152 bytes (2.1 MB) copied, 0.009481 s, 221 MB/s
[root@localhost ~]# mkfs.ext3 fs.img
mke2fs 1.40.2 (12-Jul-2007)
fs.img is not a block special device.
Proceed anyway? (y n) y

```

Figure 9: Module debugging steps in the UML instance

```

[surya@surya-server uml]$ gdb linux 8892
GNU gdb (GDB) Fedora (7.1-18.fc13)
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License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software; you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law. Type 'show copying'
and 'show warranty' for details.
This GDB was configured as "x86_64-redhat-linux-gnu".
For bug reporting instructions, please see:
<http://www.gnu.org/software/gdb/bugs/>...
Reading symbols from /home/surya/kernel/uml/linux...done.
Attaching to program: /home/surya/kernel/uml/linux, process 8892
Reading symbols from /lib64/libutil.so.1...done.
Reading symbols from /lib64/libutil.so.1
Reading symbols from /lib64/libc.so.6...done.
Loaded symbols from /lib64/libc.so.6
Reading symbols from /lib64/ld-linux-x86-64.so.2...done.
Loaded symbols from /lib64/ld-linux-x86-64.so.2
Reading symbols from /lib64/libnss_files.so.2...done.
Loaded symbols from /lib64/libnss_files.so.2
0x0000003a220a6ae0 in __nanosleep_nocancel () from /lib64/libc.so.6
Missing separate debuginfos, use: debuginfo-install glibc-2.12-1.x86_64
(gdb) add-symbol-file /home/surya/kernel/uml/
Fedora7-AMD64-root.fs linux      mods/
custom/      linux-next/
(gdb) add-symbol-file /home/surya/kernel/uml/linux-next/drivers/block/loop.ko 0x7187c000
add symbol table from file "/home/surya/kernel/uml/linux-next/drivers/block/loop.ko" at
.text_addr = 0x7187c000
(y or n) y
Reading symbols from /home/surya/kernel/uml/linux-next/drivers/block/loop.ko...done.
(gdb) p loop_unplug
$1 = {void (struct request_queue *)} 0x7187c000 <loop_unplug>
(gdb) b loop_unplug
Breakpoint 1 at 0x7187c000: file drivers/block/loop.c, line 555.
(gdb) c
Continuing.

```

Figure 10: Module debugging steps on the host system

provides access to the host system files. Once the UML system is booted, execute the following steps:

- Create a directory in the UML instance, where you will mount the host filesystem:

```
# mkdir /host
```

- Mount the host directory that contains the modules built for the UML kernel:

```
# mount none /host -t hostfs -o /code/kernel/1fy/mods
```

- Once the host directory is mounted, copy the module files to `/lib/modules` of the UML instance, with a simple `cp` command.

- Network update method (root access required): A network update involves setting up a bridge between the host and the UML system. Once the network is set up, files can be copied over the network using `scp` or an NFS share from the host, mounted in the UML system. Execute the following steps:

- On the host, you will need to have the `bridge-utils` package installed.
- You can download the source from <http://sourceforge.net/projects/bridge/files/> and compile it in your host OS. After that, run the following steps (in the same order):

```

# brctl addbr br0 (create bridge br0)
# tunctl -u 'id -u surya' (create a tap device, and
assign permissions to a normal user; replace with
username of your desired ordinary user account.)
# ifconfig eth0 0.0.0.0 promisc up (set the
system network interface and the tap interface
in promiscuous mode)
# ifconfig tap1 0.0.0.0 promisc up
# brctl addif br0 eth0 (add system interface to
the bridge)
# brctl addif br0 tap1 (add tap device to the
bridge)
# ifconfig br0 up (bring up the br0 device with
DHCP-see note below)

```

- Once the above set-up is done, the UML instance can be restarted with the following modified command line:

```
$ linux-2.6.35.rc3/linux ubda=FedoraCore6-AMD64-
root.fs mem=250M eth=vtunap, tap1
```

- As mentioned, use `scp` to copy the modules, or create an NFS share